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Quasinormal modes as a distinguisher between general relativity and $f(R)$ gravity

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Quasinormal modes (QNMs) or the ringdown phase of gravitational waves provide critical information about the structure of compact objects like black holes. Thus, QNMs can be a tool to test general relativity (GR) and possible deviations from it. In the case of GR, it has been known for a long time that a relation between two types of black hole perturbations—scalar (Zerilli) and vector (Regge-Wheeler)—leads to an equal share of emitted gravitational energy. With the direct detection of gravitational waves, it is now natural to ask whether the same relation (between scalar and vector perturbations) holds for modified gravity theories, and if not, whether one can use this as a way to probe deviations from general relativity. As a first step, we show explicitly that the above relation between Regge-Wheeler and Zerilli perturbations breaks down for a general $f(R)$ model and hence the two perturbations do not share equal amounts of emitted gravitational energy. We discuss the implication of this imbalance for observations and the no-hair conjecture.

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I. INTRODUCTION

General relativity has been highly successful and has passed all the weak-field and indirect strong-field tests [1]. The existence of black holes (BHs) is one of the key predictions of general relativity. It is argued that close to the BH singularity since the curvature is infinite, general relativity does not hold. In general, it has been suggested that at large curvatures, one needs to add higher-order curvature terms to the standard Einstein-Hilbert action [2,3]. There are several candidates for such higher-derivative additions like the contracted Riemann/Ricci curvature or higher powers of the Ricci scalar. The inclusion of such terms stabilizes the divergence structure of gravity [2].

When a star collapses to a BH or two BHs merge to form another BH, the event horizon of the remnant BH is highly distorted and radiates gravitational waves until it settles down to an equilibrium configuration [4,5]. The gravitational radiation emitted is a superposition of damped sinusoidal modes; the frequency and the damping of these quasinormal modes (QNMs) depend only on the parameters characterizing the BH (like mass, charge and angular momentum) and is independent of the initial configuration that caused the excitation [6]. Hence, QNMs play a key role, as their detection would confirm the nature of the remnant BH. The first detection of gravitational waves from the event GW150914 confirmed that the emitted gravitational waves from the remnant BH were characteristic of the QNM frequencies [7].

These recent detections [7,8] were also the first time that general relativity was directly tested in strong-field regimes [9]. This naturally raises the question of whether (and if so, how) we can use QNMs to distinguish between general relativity and modified gravity theories. For earlier works see Ref. [10]. It is important to note that QNMs provide information about the outer structure of BH spacetimes. While the introduction of the higher-derivative terms helps to cure the divergence at the curvature singularity, the higher-derivative terms can possibly change the structure of the horizon.

In the case of asymptotically flat BHs in general relativity, it is known that two kinds of gravitational perturbations exist: odd (vector) and even (scalar) [4,5]. More importantly, the two kinds of perturbations are related to each other and the net emitted gravitational energy is shared equally [5]. In this work, we check whether this equality is maintained in the modified theories of gravity. We explicitly show that the QNMs emitted from the BHs in general relativity and $f(R)$ gravity are different and obtain a quantifying tool to distinguish the same.

As mentioned earlier, there is no unique way to modify general relativity (for recent reviews, see Refs. [3,11,12]). In this work, we focus on the simplest possible extension of general relativity: the $f(R)$ model,

$$S = \frac{1}{2\kappa^2}\int d^4x\sqrt{-g}f(R), \qquad \kappa^2 = \frac{8\pi G}{c^4}. \tag{1}$$

There are two reasons for this choice. First, they are general enough that higher-order Ricci scalar terms can encapsulate high-energy modifications to general relativity, yet the

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equations of motion are simple enough that it is possible to solve them [12]. Second and most importantly, $f(R)$ theories do not suffer from Ostrogradsky instability [13]. To keep the calculations transparent, we first take $f(R) = R + \alpha R^2$, where α is a coupling constant. Later, we extend the analysis to general $f(R)$.

We use greek letters for the four-dimensional spacetime, upper case latin letters for the two angular coordinates and lower case latin letters for the two orbit coordinates. We follow the notation of Ref. [14] and set $c = G = 1$.

II. FORMALISM

We use the gauge-invariant formalism developed in Refs. [14,15]. For the four-dimensional spherically symmetric spacetime, the manifold $(\mathcal{M}, g_{\mu\nu})$ is split into two-dimensional orbit space $(\mathcal{K}^2, g_{ab})$ and a unit 2-sphere $(\mathcal{S}^2, \gamma_{AB})$. The line element is

$$ds^2 = -g(r)dt^2 + \frac{1}{g(r)}dr^2 + \rho^2(r)d\Omega^2, \tag{2}$$

where $g(r)$, $\rho(r)$ are arbitrary (continuous, differentiable) functions of the radial coordinate r. The metric perturbations about the above background (2) are given by [5,14,15]

$$h_{\mu\nu}dy^\mu dy^\nu = h_{ab}dx^a dx^b + 2h_{aB}dx^a dz^B + h_{AB}dz^A dz^B. \tag{3}$$

Note that under rotations on the 2-sphere, h_{ab}, h_{aB}, and h_{AB} transform as scalars, vectors, and tensors, respectively. Hence, one can use scalar, vector, and tensor spherical harmonics to separate the angular dependence of any field appearing in the background or the perturbed spacetime [5]. Imposing the transverse-traceless condition on the tensor perturbations, it can be shown that the tensor harmonic functions are identically zero [16]. Thus, the metric perturbations can be split into scalar and vector perturbations, i.e., $h_{\mu\nu} = h^S_{\mu\nu} + h^V_{\mu\nu}$. In the linear limit, field equations corresponding to $h^S_{\mu\nu}$ and $h^V_{\mu\nu}$ are decoupled and hence the study of the dynamics of these two types of perturbations can be handled separately [5]. For completeness, we summarize them in Appendices A and B.

From a combination of metric perturbation variables, two gauge-invariant master variables Φ^0_S and Φ^0_V for each multipole $\ell \geq 2$ can be defined that satisfy [14]

$$\frac{d^2\Phi^0_S}{dr_*^2} + (\omega^2 - V_S)\Phi^0_S = 0, \tag{4}$$

$$\frac{d^2\Phi^0_V}{dr_*^2} + (\omega^2 - V_V)\Phi^0_V = 0, \tag{5}$$

where r_* is commonly referred to as the tortoise coordinate, V_S and V_V are commonly referred to as the Regge-Wheeler and Zerilli potentials, respectively [4,5], and ω is the QNM frequency common to both modes. The following points are worth noting. First, the two master variables are not independent; they are related by [5]

$$\Phi^0_{S/V} = \frac{1}{\beta - \omega^2}\left(\mp W\Phi^0_{V/S} + \frac{d\Phi^0_{V/S}}{dr_*}\right), \tag{6}$$

where β is a function of ℓ and $W(r)$ is a function of $g(r)$ (see Appendix C). Second, the two potentials are also not independent; they are related by [5,17]

$$V_{S/V} = W(r)^2 \mp \frac{dW}{dr_*} + \beta. \tag{7}$$

The above relations hold for vacuum spacetimes and can be extended for other matter sources [17]. Third and most importantly, it can be shown that the transmission and reflection coefficients of V_S and V_V are equal for vacuum spacetimes [5]. Thus, the gravitational radiation from perturbed BHs, as detected at asymptotic spatial infinity, have equal contributions from the scalar and vector modes. Finally, the equal contributions from the scalar and vector modes may only be valid only for general relativity, and not necessarily for modified gravity models. In the rest of this article, we evaluate the contributions from these two types of modes to the $f(R)$ gravity model and show that the two contributions are not identical. Thus, QNMs provide a way to distinguish between general relativity and modified gravity models.

III. SCALAR AND VECTOR MODES IN $f(R)$

The equation of motion of Eq. (1) is

$$G_{\mu\nu} = \kappa^2 T^{\rm eff}_{\mu\nu}, \tag{8}$$

where $G_{\mu\nu}$ is the Einstein tensor, and

$$T^{\rm eff}_{\mu\nu} = \frac{\alpha}{\kappa^2}\left[2\nabla_\mu\nabla_\nu R - 2g_{\mu\nu}\Box R + \frac{1}{2}g_{\mu\nu}R^2 - 2RR_{\mu\nu}\right]. \tag{9}$$

Here, the nonlinear and higher derivatives of the Ricci curvature are bundled together into the effective energy-momentum tensor. It is important to note that a Schwarzschild BH is still the vacuum solution for the $f(R)$ model [18].

Perturbing the above field equation (8) about the vacuum solution, we get,

$$\delta G_{\mu\nu} = \delta T^{\rm eff}_{\mu\nu} \equiv 2\alpha(\nabla_\mu\nabla_\nu\delta R - g_{\mu\nu}\Box\delta R), \tag{10}$$

$$\Box\delta R - \frac{1}{6\alpha}\delta R = 0, \tag{11}$$

where Eq. (11) is the trace of Eq. (10). Before proceeding to the technical aspects, we would like to mention the following. First, unlike general relativity, gravitational radiation in $f(R)$ gravity has an extra massive scalar degree of freedom which is a mixture of longitudinal and breathing degrees of freedom [19–21]. The presence of an extra degree of freedom can be understood by performing a conformal transformation to the metric in the action (1). Under this transformation, $f(R)$ transforms to Einstein gravity plus a scalar field with a potential [22]. The dynamics of this scalar mode can be found from the trace equation (11) as a master equation akin to Eqs. (4) and (5). More specifically, by separating the angular dependence from the perturbed Ricci scalar $\delta R = \Omega(x^a)\mathbf{S}$ and making the substitution $\Omega \to \Phi = r\Omega$ [23], we obtain a modified Regge-Wheeler equation for a massive scalar field in a Schwarzschild background, i.e.,

$$\frac{d^2\Phi}{dr_*^2} + (\sigma^2 - \tilde{V}_{RW})\Phi = 0, \tag{12}$$

where $\tilde{V}_{RW} = V_{RW} + \frac{g(r)}{6\alpha}$, V_{RW} is the Regge-Wheeler potential with mass parameter $(6\alpha)^{-1}$, and $\Phi(r,t) \equiv \Phi(r)e^{i\sigma t}$. Thus, in $f(R)$, the extra mode contributes to the emitted gravitational radiation. However, if the extra mode is *not coupled* to the scalar/vector perturbations, then it is possible that the scalar and vector perturbations emitted will have an equal share, with a small percentage emitted in the extra mode. We show that this is not the case and that the extra mode couples *only* to the scalar perturbations, thus providing a *unique way* to distinguish the $f(R)$ model from general relativity.

To calculate the modification in the dynamics of scalar and vector perturbations, we need to obtain the corresponding gauge-invariant variables [14]. From Eq. (9), it is easy to note that $T^{\rm eff}_{\mu\nu}$ vanishes for vacuum spacetimes. Thus, using the Stewart-Walker lemma [24], the first-order perturbed quantity $\delta T^{\rm eff}_{\mu\nu}$ is gauge invariant. The most general perturbed stress tensor can be written as [14]

$$\delta T_{\mu\nu} = \left(\begin{array}{c|c} \tau_{ab}\mathbf{S} & r\tau_a^{(S)}\mathbf{S}_B \\ --- & ---------- \\ r\tau_a^{(S)}\mathbf{S}_B & r^2\delta P\gamma_{AB}\mathbf{S} + r^2\tau_T^{(S)}\mathbf{S}_{AB} \end{array}\right) + \left(\begin{array}{c|c} 0 & r\tau_a^{(V)}\mathbf{V}_B \\ --- & --- \\ r\tau_a^{(V)}\mathbf{S}_B & r^2\tau_T^{(V)}\mathbf{V}_{AB} \end{array}\right). \tag{13}$$

Comparing the above expression with $\delta T^{\rm eff}_{\mu\nu}$ in Eq. (10), nonvanishing quantities like the radial/angular pressure $(\tau_{ab}/\delta P)$ and anisotropic stress (τ_T) can be expressed as

$$\tau_{ab} = \frac{2\alpha}{\kappa^2}\left[D_aD_b - g_{ab}\left(\tilde{\Box} + \frac{2}{r}D^c rD_c - \frac{k^2}{r^2}\right)\right]\left(\frac{\Phi}{r}\right), \tag{14}$$

$$\tau_a^{(S)} = -\frac{2\alpha k}{\kappa^2}D_a\left(\frac{\Phi}{r^2}\right), \qquad \tau_T^{(S)} = \frac{2\alpha k^2}{\kappa^2}\frac{\Phi}{r^3}, \tag{15}$$

$$\delta P = \frac{2\alpha}{\kappa^2}\left(\frac{k^2}{2r^2} - \tilde{\Box} - \frac{2}{r}D^a rD_a\right)\left(\frac{\Phi}{r}\right),$$
$$k^2 = \ell(\ell+1), \tag{16}$$

while all other quantities vanish. This is the first key result of this work, regarding which we would like to stress a few points. First, $\tau_a^{(V)}$ and $\tau_T^{(V)}$ vanish. For $h^V_{\mu\nu}$, δR vanishes, thus implying that the vector perturbations are not modified. Second, since the radial pressure and anisotropic pressure depend on the extra mode Φ, δR for the scalar perturbations does not vanish and the dynamics get modified due to the inhomogeneous source term. (For details, see Appendix D.) Thus, the scalar and vector master equations for the spherically symmetric vacuum solution in $R + \alpha R^2$ gravity are

$$\frac{d^2\Phi_S}{dr_*^2} + (\omega^2 - V_S)\Phi_S = S_S^{\rm eff}, \tag{17}$$

$$\frac{d^2\Phi_V}{dr_*^2} + (\omega^2 - V_V)\Phi_V = 0.. \tag{18}$$

Defining $H(r) \equiv H = k^2 + \frac{6M}{r} - 2$ and $g(r) \equiv g = 1 - \frac{2M}{r}$ we have,

$$\Phi_V = \Phi_V^0; \qquad \Phi_S = \Phi_S^0 + \tilde{\Phi}_S, \qquad \tilde{\Phi}_S = -\frac{4\alpha}{H}\Phi, \tag{19}$$

$$S_S^{\rm eff} = \left[C_1(\sigma,\omega,r) + C_2(\sigma,\omega,r)\frac{d}{dr_*}\right]\tilde{\Phi}_S \tag{20}$$

the details of which are given in Appendix E. The coefficients C_1 and C_2 are given by

$$\begin{aligned} C_1(\sigma,\omega,r) = {} & \sigma^2\left(1+\frac{\sigma}{\omega}\right) - \frac{Mg}{r^3}\left(1+\frac{18M}{rH}\right) \\ & - \left(\frac{\sigma}{\omega}\right)\frac{g}{r^2}\left[\frac{54M^2}{r^2H} - \frac{72gM^2}{r^2H^2} - \frac{18M}{rH}\right. \\ & \left. + \frac{1}{2}\frac{P_1}{H} - \frac{3M}{r} + \frac{\tilde{V}_{RW}}{g}\right], \end{aligned} \tag{21}$$

$$C_2(\sigma,\omega,r) = \frac{3M}{r^2} - \left(\frac{\sigma}{\omega}\right)\left[\frac{12Mg}{r^2H} - \frac{M}{r^2}\right];$$
$$P_1 = -\frac{48M^2}{r^2} + \frac{8M}{r}(8-k^2) - 2k^2(k^2-2). \tag{22}$$

Third, Eqs. (17) and (18) along with Eq. (12) form the complete set of equations describing the three propagating degrees of freedom. Finally, due to the change of Φ_S in Eq. (19), the isospectral relation (6) is no longer valid [although Eq. (7) still holds]. Physically this can be understood as the scalar mode leaking energy to the extra mode Φ via the source term (17). Hence, there is an imbalance between the contributions of the scalar- and vector-type modes to the emitted gravitational radiation.

The above analysis can be extended for a general $f(R) = \sum_{n=1}^{\infty} a_n R^n$ model where the effective stress tensor is [12]

$$T^{\rm eff}_{\mu\nu} = \frac{1}{\kappa^2 f'}\left[\nabla_\mu\nabla_\nu f' + \frac{g_{\mu\nu}}{2}(f - Rf') - g_{\mu\nu}\Box f'\right], \quad (23)$$

and $f'(R) = df(R)/dR$. About $R = 0$, the perturbed stress tensor becomes

$$\delta T^{\rm eff}_{\mu\nu} = \frac{1}{\kappa^2}(\nabla_\mu\nabla_\nu\delta f' - g_{\mu\nu}\Box\delta f') \quad (24)$$

where $\delta f' = 2\alpha\delta R$. Substituting $\delta f'$ in Eq. (24) leads to a perturbed stress that is identical to Eq. (10). Thus, all the results obtained for $R + \alpha R^2$ gravity are valid for general $f(R)$.

IV. DISTINGUISHING $f(R)$ FROM GR

To go about distinguishing general relativity and $f(R)$ gravity, we need to know $S^{\rm eff}_S$ for which we have to obtain the form of the extra mode Φ. To obtain Φ, we need to understand the form of the modified Regge-Wheeler potential:

$$\tilde{V}_{RW}(\tilde{r}) = \left(1 - \frac{2}{\tilde{r}}\right)\left[\left(\frac{\ell(\ell+1)}{\tilde{r}^2} + \frac{2}{\tilde{r}^3}\right)\frac{1}{M^2} + \frac{1}{6\alpha}\right], \quad (25)$$

where $\tilde{r} = \frac{r}{M}$ is a dimensionless radial coordinate. The Eöt-Wash experiments provide a bound on α: $\alpha \leq 10^{-9}\ m^2$ [20,25]. This is the default value used in this study. This leads to a large value of $\tilde{V}_{RW}$ at infinity, leading to a large potential barrier for the extra mode. Hence, Φ does not propagate to ∞, but falls back into the BH after excitation [unless $\sigma^2 \geq (6\alpha)^{-1}$, which becomes the cutoff frequency for the extra mode excitation [20,23]], i.e. a purely decaying solution both in r and t. For astrophysical-sized BHs and $\alpha \sim 10^{-9}$ m^2, Eq. (25) can be approximated as

$$\tilde{V}_{RW} \approx \left(1 - \frac{2}{\tilde{r}}\right)\frac{1}{6\alpha} \quad (26)$$

which indicates that for observed BH candidates the potential is almost independent of the mass. For small BHs ($\sim 10^{-5}\ M_\odot$), the local maxima reappears (as seen in Fig. 1), and the problem becomes a scattering one. In the next section we will discuss the potential significance of this result for the no-hair theorem [26].

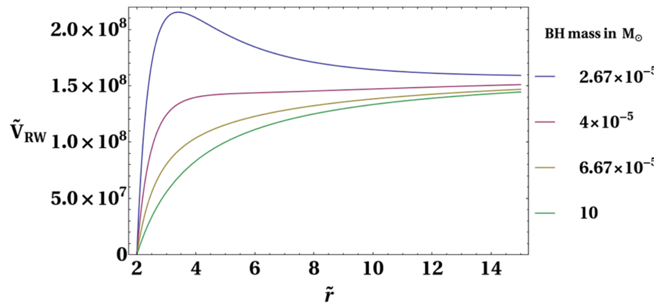

FIG. 1. Plot of Eq. (25) for a range of BH masses. For small M ($\sim 10^{-5}\ M_\odot$), the local maxima reappears, making it similar to the scattering potentials $V_{S/V}$. For such small masses (which is only possible for primordial BHs), the massive scalar radiation can propagate to ∞ for $\sigma^2 \geq (6\alpha)^{-1}$, and can have a larger share of the net emitted gravitational radiation. For comparison, the green curve shows the potential encountered by Φ for a 10 $M_\odot$ BH.

An estimate of the maximum deviation from GR can be made by considering the assumption that the intensity of a field (massless or massive) at a point depends inversely on the strength of the potential at that point. Given this, we can define a ratio of the intensities of the extra mode (I_{MS}) and the scalar perturbation (I_S) of GR, at a point outside the horizon as

$$\frac{I_{MS}}{I_S} = \frac{V_S}{\tilde{V}_{RW}} = \frac{6\alpha}{M^2}\left[\frac{\ell(\ell+1)}{\tilde{r}^2} - \frac{6}{\tilde{r}^3}\right], \quad (27)$$

where, $\tilde{r} = \frac{r}{M}$. Figure 2 shows a numerical estimate of the relative intensity of the extra mode to the scalar perturbation, at a distance $\tilde{r}M$ from the horizon. Hence, it is also an upper limit of the relative difference in intensities of the scalar- and vector-type perturbations (which is zero in GR). It is important to note that V_V is used instead of V_S because of the isospectrality relation between V_V and V_S [Eq. (7)]. For 10 $M_\odot$ BH, $\ell = 2$, and at a distance of $100M$ from the BH, the relative difference is $\leq 10^{-14}$, while total deviation

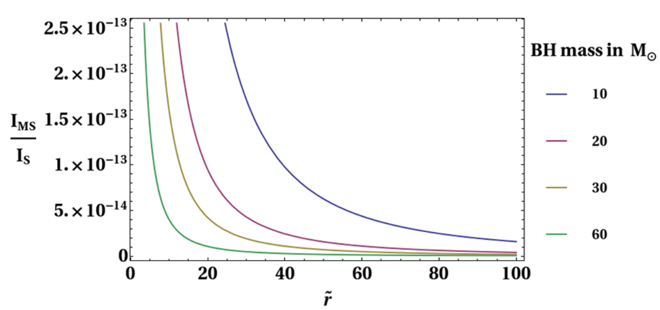

FIG. 2. The ratio of the potentials is plotted against the dimensional radius $\tilde{r}$ for different values of M and $\ell = 2$. The relative intensity reaches a maximum close to the horizon since the extra mode is localized there, and quickly decreases.

can be obtained by integrating from the horizon to infinity i.e. $\sim 10^{-10}$.

V. CONCLUSIONS AND DISCUSSIONS

Motivated by the historic direct detection of gravitational waves, we have investigated the following question: how could we use QNMs to distinguish between general relativity and modified gravity theories? We have explicitly shown that while the isospectral relations between scalar and vector perturbations hold for general relativity, they *do not hold* for $f(R)$ gravity. We have shown that the vector perturbations remain unchanged while the scalar perturbations are modified by the appearance of an extra mode as the source term.

Physically, the modifications to the scalar perturbations appear by treating the extra mode as an effective stress tensor of a perturbed fluid that vanishes in the background spacetime. The perturbed quantities (14)–(15) in turn make up an effective source term which couples to the scalar-type master equation and modifies the dynamics. The nature of the effective potential (25) makes the dynamics of the extra mode different from the scalar and vector perturbations. In other words, while the scalar and vector perturbations escape to ∞ as gravitational radiation, the extra mode does not.

The net emitted gravitational energy in $f(R)$ gravity gets shared among the three (scalar, vector and the extra mode) perturbations instead of two perturbations in general relativity. While the vector type retains its contribution, Φ takes away some of the energy released through the scalar type. This in turn also decreases the net gravitational energy that propagates to infinity, owing to the massive nature of the extra mode. The relative intensity of Φ_S and Φ at any point in the exterior spacetime was found to be capped above by α, indicating the order of the maximum fractional change in energy. At a large, fixed distance $(100M)$ from a 10 solar mass BH, we found that the intensity of Φ relative to Φ_S is of the order of 10^{-14}.

The analysis naturally leads to the following question: how can this analysis can be used in the current and future gravitational-wave detectors to distinguish between general relativity and $f(R)$? The change in the intensities of Φ_S and Φ_V can be found from the observations, since a combination of the two manifests as a $+$ and $\times$ polarization at asymptotic ∞ [27,28],

$$h_+ = \frac{1}{r}\sum_l [\Phi_S^{(\ell)}\mathbf{S}_{\theta\theta}^{(\ell)} + \Phi_V^{(\ell)}\mathbf{V}_{\theta\theta}^{(\ell)}], \tag{28}$$

$$h_\times = \frac{1}{r\sin\theta}\sum_\ell [\Phi_S^{(\ell)}\mathbf{S}_{\theta\phi}^{(\ell)} + \Phi_V^{(\ell)}\mathbf{V}_{\theta\phi}^{(\ell)}], \tag{29}$$

the details of which are included in Appendix F. Using the orthogonality relations of the scalar and vector harmonics [29], the above expressions can be rewritten as

$$\frac{\Phi_S^{(\ell)}}{r} = \int d\Omega h_+ \mathbf{S}_{\theta\theta}^{(\ell)} + \int d\Omega \sin\theta h_\times \mathbf{S}_{\theta\phi}^{(\ell)}, \tag{30}$$

$$\frac{\Phi_V^{(\ell)}}{r} = \int d\Omega h_+ \mathbf{V}_{\theta\theta}^{(\ell)} + \int d\Omega \sin\theta h_\times \mathbf{V}_{\theta\phi}^{(\ell)}, \tag{31}$$

where the integration is carried over the 2-sphere which projects out $\Phi_{S/V}$ (for each multipole ℓ) from the detected polarization. A network of laser interferometers and more detections will be able to measure the individual intensity and angular dependence respectively, of these two polarizations, and hence any difference in the intensities of Eqs. (30) and (31) (and hence, the difference in the radiated energy between the two modes) can be calculated.

In Ref. [30], three model-independent tests have been proposed: we are currently investigating a way to use these tests for the $f(R)$ model. Our present analysis is in line with the results of Ref. [31] where it was shown that modifications to gravity could hide in the statistical indeterminacies of parameter matching. In other words, a large class of non-Einstein black holes can mimic the quasinormal spectra of black holes in general relativity. Our analysis is independent of the quasinormal spectra, and the energy shares of the two polarizations of gravitational waves can distinguish between general relativity and $f(R)$ gravity. We are also currently extending the analysis for rotating and charged black holes in $f(R)$.

The issue of the broken balance between scalar and vector modes is not unique to $f(R)$ theories, and hence other modified theories of gravity will show the effect studied in this paper. Moreover, the presence of other sources like matter and electromagnetic fields [5,32] will also contribute to breaking the balance between the two modes of radiation, with the imbalance depending on the distribution of such matter/fields. In general relativity, the energetic difference can be treated as a gravitational measure of *matter+fields* between the source and the detectors. This can be understood by noting that the inhomogeneous source terms appear differentially for general relativity and for the $f(R)$ model [see Eqs. (17)–(18) and the rhs of Eqs. (4)–(5)]. However, sky observations give us an electromagnetic measure of the *matter and the fields*. Hence, a comparison between the electromagnetic and gravitational measures of matter should, in principle, help us put tighter constraints than Solar System tests on the parameter α, and consequently, any deviation from general relativity.

Finally, it is important to note that both in general relativity and $f(R)$ gravity, the different perturbations have potentials with no local minima. Specifically, in the case of the extra scalar mode, Φ can either fall back into the BH or propagate to infinity. It cannot exist in a stable configuration around the BH, which is an indication of the *no scalar hair* [26] property of BH spacetimes. Thus, the uniqueness theorem of general relativity holds true for $f(R)$ and Lovelock gravity theories in higher dimensions [33].

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APPENDIX A: SCALAR, VECTOR, AND TENSOR HARMONICS

Any field appearing in the background or the perturbed spacetime can be classified as a scalar, vector or tensor depending on how they transform under rotations in the spherically symmetric spacetime. Thus, the angular dependence of any object is separated out using spherical harmonics of three types, which are further subdivided as follows:

(1) The angular dependence of a scalar function is expressed as a sum of scalar spherical harmonics $\mathbf{S} \propto Y_{lm}(\theta, \phi)$.

(2) An arbitrary vector field U_A is expressed as $U_A = V_A + D_A S$, where S is a scalar, corresponding to a $\mathbf{S}_A$ (*scalar-type vector harmonic*) and a $\mathbf{V}_A$ (*vector harmonic function*).

(3) An arbitrary tensor field X_{AB} is expressed as $X_{AB} = T_{AB} + 2D_{(A}V_{B)} + \hat{L}_{AB}S + g_{AB}S$, where g_{AB} is the metric on a two-dimensional subspace. T_{AB} is transverse-traceless tensor field, V_A is a transverse vector field, the projection operator $\hat{L}_{AB} \equiv D_A D_B - \frac{1}{2} g_{AB}$ obtains a traceless tensor field from a scalar, and $g_{AB}S$ is the trace part of X_{AB}. These correspond to $\mathbf{T}_{AB}$ (*tensor harmonic function*), $\mathbf{V}_{AB}$ (*vector-type harmonic tensor*), $\mathbf{S}_{AB}$ (*scalar-type harmonic tensor*), and $\gamma_{AB}\mathbf{S}$.

Out of the seven possibilities listed above, the scalar, vector, and tensor harmonic functions will be called "pure" and the remaining four will be called "derived." The three pure harmonic functions satisfy the following eigenvalue equations:

$$(\hat{\Box} + k^2)\mathbf{S} = 0,$$
$$(\hat{\Box} + k^2)\mathbf{V}_A = 0,$$
$$(\hat{\Box} + k^2)\mathbf{T}_{AB} = 0, \tag{A1}$$

where $\mathbf{S}$, $\mathbf{V}_A$, and $\mathbf{T}_{AB}$ are scalar, vector, and tensor harmonic functions respectively, and $k^2 = \ell(\ell+1)$.

A vector quantity that can be defined from a pure harmonic function is the scalar-type vector harmonic, defined as

$$\mathbf{S}_A = -\frac{1}{k} D_A \mathbf{S}. \tag{A2}$$

$\mathbf{S}_A$ and $\mathbf{V}_A$ form the basis for the angular dependency of components transforming as vectors. Similarly, two tensor quantities can be derived which are the scalar- and vector-type tensor harmonic functions

$$\mathbf{S}_{AB} = \frac{1}{k^2} D_A D_B \mathbf{S} + \frac{1}{2}\gamma_{AB}\mathbf{S}, \tag{A3}$$

$$\mathbf{V}_{AB} = -\frac{1}{2k}(D_A \mathbf{V}_B + D_B \mathbf{V}_A). \tag{A4}$$

Equations (A3) and (A4) along with $\mathbf{T}_{AB}$ and $\gamma_{AB}\mathbf{S}$ form the basis for the angular dependence of any fields transforming as tensors.

APPENDIX B: VANISHING OF TENSOR PERTURBATIONS ON A UNIT 2-SPHERE, AND ITS PRESENCE ON A 3-SPHERE

1. Trivial $h^{(T)}_{\mu\nu}$ on S^2

The S^2 background metric is given by

$$g_{AB} \equiv \begin{pmatrix} 1 & 0 \\ 0 & \sin^2\theta \end{pmatrix}, \tag{B1}$$

on which a transverse and traceless tensor perturbation has the form

$$h_{AB} \equiv \begin{pmatrix} -\alpha(x^a,\theta,\phi) & \beta(x^a,\theta,\phi) \\ \beta(x^a,\theta,\phi) & \alpha(x^a,\theta,\phi)\sin^2\theta \end{pmatrix} \tag{B2}$$

and satisfies $D^A h_{AB} = 0$, which leads to two equations:

$$\partial_\phi(\beta\sin\theta) - \sin\theta\partial_\theta(\alpha\sin^2\theta) = 0, \tag{B3}$$

$$\partial_\theta(\beta\sin\theta) + \sin\theta\partial_\phi\alpha = 0. \tag{B4}$$

These two lead to the following conditions:

$$\hat{\Box}(\alpha\sin^2\theta) = 0, \tag{B5}$$

$$\hat{\Box}(\beta\sin\theta) = 0. \tag{B6}$$

Hence $\alpha\sin^2\theta$ and $\beta\sin\theta$ must be proportional to the $\ell = 0$ harmonic function, which is a constant. Then α and β should be of the form

$$\alpha = \frac{f(x^a)}{\sin^2\theta}, \tag{B7}$$

$$\beta = \frac{g(x^a)}{\sin\theta}. \tag{B8}$$

For α and β to be regular at the poles ($\theta = 0$), $f(r)$ and $g(r)$ need to go to zero, implying that tensor perturbations on S^2 vanish.

2. Nontrivial $h_{\mu\nu}^{(T)}$ on S^3

The transverse-traceless tensor perturbation on a unit 3-sphere can be written as follows:

$$h_{AB} \equiv \begin{pmatrix} -\alpha(\psi,\theta,\phi) & \alpha_1(\psi,\theta,\phi) & \alpha_2(\psi,\theta,\phi) \\ \alpha_1(\psi,\theta,\phi) & \alpha(\psi,\theta,\phi)\sin^2\psi - \beta(\psi,\theta,\phi)\sin^2\psi & \beta_1(\psi,\theta,\phi) \\ \alpha_2(\psi,\theta,\phi) & \beta_1(\psi,\theta,\phi) & \beta(\psi,\theta,\phi)\sin^2\psi\sin^2\theta \end{pmatrix} \tag{B9}$$

with a background metric

$$g_{AB} \equiv \begin{pmatrix} 1 & 0 & 0 \\ 0 & \sin^2\psi & 0 \\ 0 & 0 & \sin^2\psi\sin^2\theta \end{pmatrix}. \tag{B10}$$

The perturbed metric has five functions and three equations, whereas for the 2-sphere case the number of functions and equations were equal. $D^A h_{AB} = 0$ corresponds to the following three equations:

$$-\sin\psi\sin^2\theta\partial_\psi(\alpha\sin^3\psi) + \sin^2\psi(\sin^2\theta\partial_\theta\alpha_1 + \sin\theta\cos\theta\alpha_1) + \partial_\phi(\alpha_2\sin^2\psi) = 0, \tag{B11}$$

$$\sin^2\theta\partial_\psi(\alpha_1\sin^2\psi) - \sin^2\psi\partial_\theta(\beta\sin^2\theta) + \sin^2\psi(\sin^2\theta\partial_\theta\alpha + \sin\theta\cos\theta\alpha) = 0, \tag{B12}$$

$$\sin\theta\partial_\psi(\alpha_2\sin^2\psi) + \partial_\theta(\beta_1\sin\theta) + \sin\theta\sin^2\psi\partial_\phi\beta = 0. \tag{B13}$$

α_1 and α_2 do not have a derivative-of-a-product form in all three equations. All five functions cannot be written individually in the $\hat{\Box} f = 0$ form. At most, three such relations are possible. This shows how tensor perturbations are present in $1+3$ (cosmology) but not in $2+2$ (black holes).

APPENDIX C: ISOSPECTRALITY

The authors of Ref. [5] found that the potentials V_S and V_V have the same reflection and transmission coefficients, which implies that the net emitted gravitational radiation has equal contributions from scalar and vector perturbations. The authors of Ref. [5] showed that if the two potentials are related to each other through

$$V_{S/V} = W^2 \mp \frac{dW}{dr_*} + \beta, \tag{C1}$$

then the master variables Φ_S and Φ_V are related as

$$\Phi_{S/V} = \frac{1}{\beta - \omega^2}\left(\mp W\Phi_{V/S} + \frac{d\Phi_{V/S}}{dr_*}\right). \tag{C2}$$

$W(r)$ and β are given by

$$W(r) = \frac{6M(2M-r)}{r^2(6M+2\lambda r)} - \frac{\lambda(\lambda+1)}{3M}, \tag{C3}$$

$$\beta = -\frac{\lambda^2(\lambda+1)^2}{9M^2}, \tag{C4}$$

where $\lambda = \frac{(\ell-1)(\ell+2)}{2}$. The above relations can be generalized to spacetimes with a cosmological constant (Λ) or electromagnetic fields (as in the case of a Reissner-Nordström BH) [5], although they do not hold in general for arbitrary $T_{\mu\nu}$.

Using Eq. (C2) and proper boundary conditions (purely ingoing at the horizon, purely outgoing at ∞), if ω is a QNM frequency of one type, it is also the QNM frequency of the other type. This implies that the scalar and vector perturbations also share the same spectra.

APPENDIX D: RELATING HIGHER-DERIVATIVE TERMS WITH THE PERTURBED ENERGY-MOMENTUM TENSOR

The Christoffel symbols of the background metric written in $2+2$ form become

$$\Gamma^a_{bc} = {}^2\Gamma^a_{bc}, \tag{D1}$$

$$\Gamma^a_{BC} = -rD^a r\gamma_{BC}, \tag{D2}$$

$$\Gamma^A_{aB} = \frac{D_a r}{r}\delta^A_B, \tag{D3}$$

$$\Gamma^A_{BC} = \hat{\Gamma}^A_{BC} \tag{D4}$$

where ${}^2\Gamma^a_{bc}$ and $\hat{\Gamma}^A_{BC}$ are Christoffel symbols on $\mathcal{K}^2$ and $\mathcal{S}^2$ respectively. Using Eqs. (D1)–(D4), various double covariant derivatives for some scalar function $F(y^\mu)$ are calculated as follows:

$$\Box F = \tilde{\Box} F + \frac{1}{r^2}\hat{\Box} F + \frac{2}{r} D^a r D_a F, \tag{D5}$$

$$\nabla_a \nabla_b F = D_a D_b F, \tag{D6}$$

$$\nabla_a \nabla_B F = r D_a \left(\frac{1}{r} D_B F\right), \tag{D7}$$

$$\nabla_A \nabla_B F = D_A D_B F + r D^a r D_a F \gamma_{AB}. \tag{D8}$$

The higher-derivative terms of the modified field equation, bundled as an effective energy-momentum tensor have a a perturbation around $R = 0$ given by

$$\delta T^{\rm eff}_{\mu\nu} = \frac{2\alpha}{\kappa^2}(\nabla_\mu \nabla_\nu \delta R - g_{\mu\nu}\Box \delta R), \tag{D9}$$

Since δR is a scalar function, its angular dependence is separated by

$$\delta R = \Omega(x^a)\mathbf{S}(z^A) \tag{D10}$$

Using Eqs. (D5)–(D8) in Eq. (D9) and using definitions (A2) and (A3) the following relations are obtained:

$$\delta T^{\rm eff}_{ab} = \frac{2\alpha}{\kappa^2}\left[D_a D_b \Omega - g_{ab}\left(\tilde{\Box}\Omega + \frac{2}{r} D^c r D_c \Omega - \frac{k^2}{r^2}\Omega\right)\right]\mathbf{S}, \tag{D11}$$

$$\delta T^{\rm eff}_{aB} = -\frac{2\alpha}{\kappa^2} k r D_a \left(\frac{\Omega}{r}\right)\mathbf{S}_B, \tag{D12}$$

$$\delta T^{\rm eff}_{AB} = \frac{2\alpha}{\kappa^2}\left[k^2 \Omega \mathbf{S}_{AB} - r^2 \gamma_{AB}\left(\tilde{\Box}\Omega + \frac{2}{r} D^a r D_a \Omega - \frac{k^2}{2r^2}\Omega\right)\mathbf{S}\right]. \tag{D13}$$

τ_{ab}, $\tau_a^{(S/V)}$, δP, and $\tau_T^{(S/V)}$ are found by comparing the above relations with the perturbed energy-momentum tensor (13)

$$\tau_{ab} = \frac{2\alpha}{\kappa^2}\left[D_a D_b - g_{ab}\left(\tilde{\Box} + \frac{2}{r} D^c r D_c - \frac{k^2}{r^2}\right)\right]\left(\frac{\Phi}{r}\right), \tag{D14}$$

$$\tau_a^{(S)} = -\frac{2\alpha k}{\kappa^2} D_a \left(\frac{\Phi}{r^2}\right), \tag{D15}$$

$$\delta P = \frac{2\alpha}{\kappa^2}\left(\frac{k^2}{2r^2} - \tilde{\Box} - \frac{2}{r} D^a r D_a\right)\left(\frac{\Phi}{r}\right), \tag{D16}$$

$$\tau_T^{(S)} = \frac{2\alpha k^2}{\kappa^2}\frac{\Phi}{r^3}, \tag{D17}$$

$$\tau_a^{(V)} = 0, \tag{D18}$$

$$\tau_T^{(V)} = 0 \tag{D19}$$

where $\Phi = r\Omega$ is the extra scalar mode.

APPENDIX E: THE EFFECTIVE SOURCE TERM

The inhomogeneous source term for a general matter perturbation in a Schwarzschild background was obtained for $m + n$ spacetimes with an electromagnetic presence in Ref. [14]. Here, we use a restricted version of the above by putting the background and perturbed electromagnetic sources to zero and $m = n = 2$. We obtain,

$$S_S^{\rm eff} = \frac{g}{rH}\left[-HS_T - \frac{P_1}{H}\frac{S_t}{i\omega} - 4g\frac{r(S_t)'}{i\omega} - 4rgS_r + \frac{P_2}{H}\frac{rS_t^r}{i\omega} + 2r^2\frac{(S_t^r)'}{i\omega} + 2r^2 S_r^r\right], \tag{E1}$$

where the prime denotes a radial derivative and

$$S_b^a = \kappa^2 \tau_b^a, \qquad S_a = \frac{r\kappa^2}{k}\tau_a^{(S)}, \qquad S_T = \frac{2r^2\kappa^2}{k^2}\tau_T^{(S)}, \tag{E2}$$

$$P_1 = -\frac{48M^2}{r^2} + \frac{8M}{r}(8 - k^2) - 2k^2(k^2 - 2), \tag{E3}$$

$$P_2 = \frac{24M}{r}, \qquad H = k^2 + \frac{6M}{r} - 2. \tag{E4}$$

Equation (E2) was calculated by using Eqs. (D14), (D15), and (D17) and substituting it into (E1). The time dependence of Φ is separated out using

$$\Phi(r,t) \equiv \Phi(r)e^{i\sigma t}. \tag{E5}$$

Double radial derivatives are reduced by using the equation of motion of Φ

$$\frac{d^2\Phi}{dr_*^2} + (\sigma^2 - \tilde{V}_{RW})\Phi = 0 \tag{E6}$$

from which the effective source term is obtained as

$$S_S^{\rm eff} = \left[C_1(\sigma,\omega,r) + C_2(\sigma,\omega,r)\frac{d}{dr_*}\right]\tilde{\Phi}_S \tag{E7}$$

where $\tilde{\Phi}_S$ is defined in Eq. (19) and the coefficients are obtained as

$$C_1(\sigma,\omega,r) = \sigma^2\left(1+\frac{\sigma}{\omega}\right) - \frac{Mg}{r^3}\left(1+\frac{18M}{rH}\right) - \left(\frac{\sigma}{\omega}\right)\frac{g}{r^2}\left[\frac{54M^2}{r^2H} - \frac{72gM^2}{r^2H^2} - \frac{18M}{rH} + \frac{1}{2}\frac{P_1}{H} - \frac{3M}{r} + \frac{\tilde{V}_{RW}}{g}\right], \tag{E8}$$

$$C_2(\sigma,\omega,r) = \frac{3M}{r^2} - \left(\frac{\sigma}{\omega}\right)\left[\frac{12Mg}{r^2H} - \frac{M}{r^2}\right];$$
$$P_1 = -\frac{48M^2}{r^2} + \frac{8M}{r}(8-k^2) - 2k^2(k^2-2). \tag{E9}$$

APPENDIX F: RADIATION AT INFINITY

In order to connect the scalar/vector master variables to the radiation detectable at asymptotic spatial infinity, and hence the polarizations, the asymptotic behavior of the perturbed metric was studied in Ref. [27] and found to be

$$\lim_{r\to\infty} h_{\mu\nu} \sim \left(\begin{array}{c|c} O\left(\frac{1}{r}\right) & O(r^0) \\ --- & --- \\ O(r^0) & O(r) \end{array}\right). \tag{F1}$$

Thus, the leading-order contribution to gravitational radiation comes only from the h_{AB} components of the perturbed metric. h_{AB} split into scalar and vector perturbations have the following forms:

$$h^V_{AB} = 2r^2 H^V_T \mathbf{V}_{AB}, \tag{F2}$$

$$h^S_{AB} = 2r^2(H_L \mathbf{S}\gamma_{AB} + H^S_T \mathbf{S}_{AB}), \tag{F3}$$

where $\mathbf{S}, \mathbf{S}_{AB}$, and $\mathbf{V}_{AB}$ are defined in Appendix A.

Equation (F1) in a radiation gauge [27,28] should correspond to the transverse-traceless metric of the Minkowski perturbation, from which it can be inferred that only the transverse components of h_{AB} contribute at leading order at infinity [27,28]. Thus, only H^V_T and H^S_T contribute to gravitational radiation. This can be seen as follows: using the definitions of gauge-invariant variables in Ref. [14] and enforcing that all other metric components and sources die off faster than $H^{S/V}_T$, from the definition of master variables and the perturbed field equations, at large r,

$$\Box H^{S/V}_T = 0 \tag{F4}$$

In Refs. [27,28], the authors obtained the relations between the metric perturbation and the master variables as

$$H^S_T \sim \frac{\Phi^0_S}{r}, \tag{F5}$$

$$H^V_T \sim \frac{\Phi^0_V}{2r}. \tag{F6}$$

From the TT metric of the Minkowski perturbation, the polarizations can be identified as

$$h_+ \equiv \frac{h_{\theta\theta/\phi\phi}}{r^2}, \tag{F7}$$

$$h_\times \equiv \frac{h_{\theta\phi}}{r^2\sin\theta}. \tag{F8}$$

Applying the TT condition to Eqs. (F2) and (F3), and using Eqs. (F5)–(F8) we obtain relations between the scalar/vector master variable and the polarizations at each multipole ℓ as

$$h^{(\ell)}_+ \sim \frac{1}{r}[\Phi^{0(\ell)}_S \mathbf{S}^{(\ell)}_{\theta\theta} + \Phi^{0(\ell)}_V \mathbf{V}^{(\ell)}_{\theta\theta}], \tag{F9}$$

$$h^{(\ell)}_\times \sim \frac{1}{r\sin\theta}[\Phi^{0(\ell)}_S \mathbf{S}^{(\ell)}_{\theta\phi} + \Phi^{0(\ell)}_V \mathbf{V}^{(\ell)}_{\theta\phi}]. \tag{F10}$$

Since the extra scalar mode does not travel to ∞, Eqs. (F9) and (F10) hold for $f(R)$ gravity as well, with Φ^0_S replaced by the modified Φ_S, whose intensity at the detector is different from Φ^0_V.

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